

TEJAL BEDMUTHA

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EDUCATION

Columbia University, Sep 2025 - Dec 2027
Masters in Computer Science New York, United States.
Veermata Jijabai Technological Institute (VJTI), Aug 2019 - Jul 2023
Bachelor of Textile Technology and Manufacturing, GPA- 3.95/4 Mumbai, India.
Ranked first in the program, All Rounder Best Student of the Batch 2019-2023, Alumni Association Scholarship.

PUBLICATIONS

- **A Survey on Sensor-based Planning and Control for Unmanned Underwater Vehicles.**
Tejal Bedmutha & Shivam Vishwakarma. (Submitted to Wiley Journal of Field Robotics(2025)).
- **Instantaneous Planning, Control & Safety for Navigation in Unknown Underwater Spaces.**
Veejay Karthik, Udit Ekansh, Tejal Bedmutha, Rohan Deshpande, Shivam Vishwakarma. (Submitted to IEEE Transactions on Robotics and Automation(2025)).
- **Software Architecture of the Navigation Stack for Simulation of an Autonomous Underwater Vehicle.**
Tejal Bedmutha & Shivam Vishwakarma. (Submitted to IEEE Robotics and Automation Magazine(2025)).
- **Comparative Analysis and Optimization of Offline UAV Clustering Algorithms**
Tejal Bedmutha , Jatin Salve. (Under Preparation). [Link](#)
- **Comparative study of in-situ polymerisation by polyaniline on different fabrics for development of conductive fibres**
Tejal Bedmutha, Jivita Gorad, Ritik Nawgire, Aditya Jadhav.(Undergrad Thesis). [Link](#)

WORK EXPERIENCE

Indian Institute of Technology, Bombay May 2024 - Jun 2025
Research Assistant(Project Lead). Mumbai, India.

- **Modeled and integrated** SONAR sensor into a simulation environment, leveraging the Octomap library to generate precise octomaps from point clouds with **92%** accuracy, accounting for sensor noise and distortions.
- **Formulated and Tested** dynamic mapping functionality to classify static and dynamic obstacles, achieving **90%** accuracy.
- **Led and Devised** a navigation system leveraging sensor data for dynamic mapping and path planning, attaining **approx 30m** target accuracy, and deployed it for vehicle testing.

MIKO.AI Jul 2023 - Apr 2024
Robotics Engineer. Mumbai, India.

- **Led development** of a 1-DOF interactive robot by designing the software architecture, implementing utility functions, and collaborating with cross-functional teams to enable seamless real-time interactions with **99%** collision and operational safety.
- **Engineered** a dual-detection system using 1D LiDAR, classifying edges and obstacles using a gradient-based algorithm, attaining **99%** accuracy and minimizing false positives to **0.8%**, ensuring improved obstacle navigation.
- **Designed** the LED control module for the Miko Mini, integrating advanced sensor fusion and system modelling techniques, enhancing interactive responses and user experience.

MITACS - GRI | MATLAB, Gazebo, Simulink. May 2022 - Aug 2022
Research Intern. Montreal, Canada.

- **Conducted a comprehensive literature review** on various evolutionary algorithms for clustering in scheduling-based applications.
- **Devised** an efficient clustering algorithm for UAVs, reducing clustering time by **20 seconds** compared to conventional evolutionary methods, and optimizing pre-takeoff processes for faster deployment.

PROJECTS

Geometric Motion Planning | Python (Numpy, PyTorch). [Link](#) May 2024 - Oct 2024
Developed an open-source 360° LiDAR simulator and applied it for 3D motion planning of ground robots

- **Built** a Python-based LiDAR simulator capable of generating 360° environmental data.
- **Devised** a geometry-driven algorithm for 3D motion planning of robots in dense, dynamic settings.

TECHNICAL SKILLS

OS : Windows, Linux, Mac-OS, ROS/ROS2(Meta-OS).
Languages : C++, Python, C, Embedded-C, MATLAB, Javascript, HTML/CSS.
Databases : Shell Scripting, MySQL.
Technologies/Frameworks : OpenCV, Scikit-Learn, Numpy, Git.
Software : Gazebo, Coppeliassim, Simulink, MATLAB, MS Excel.
Relevant Coursework : Linear Algebra, Probability, Discrete Mathematics, Systems and Control Theory.